

Fuzzy Clock: Decision-Specific Timing Information

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August 2026

Abstract

Economic responses indexed by recorded dates require event-time validation tailored to the specified decision. Let latent event time G be recorded as $W = G + U$, and let m denote a response indexed by W . For a finite linear action set, common adjoint bridges for pairwise payoff contrasts identify the action ordering within a clock-summary cell. For a nonnegative polynomial timing loss of even degree d , the recorded response and clock moments through d recover the scheduled-loss function. An adjacent-moment construction preserves m and clock moments through $d - 1$ and changes minimum scheduled loss and the policy choice over an interval of flexibility costs. With quadratic loss, mean displacement shifts the verified-origin scheduled date, and displacement variance determines the gap between recorded-origin and verified-origin scheduled loss. An application to European gas disruptions estimates a three-week network response around the June 2022 Nord Stream reduction, combines it with regulator and physical timing records, and maps the available clock information into scheduling and storage-option benchmarks.

JEL codes: C14, C18, C21, C32, F51, Q41. *Keywords:* event time, event study, treatment timing, inverse problem, counterfactual identification, partial identification, natural gas.

1. Introduction

Economic decisions often use dates supplied by records: a notification can open a reserve call, a certification can start a transfer, and a detected disruption can start procurement. The underlying economic event may begin at a different time. The required validation information depends on how that difference changes the response used in the decision.

The decision determines which clock information is relevant. A scheduling date depends on mean displacement. A flexibility choice under quadratic mismatch also depends on displacement variance, and a finite action ordering depends on bridges for its payoff contrasts. Three steps connect these objects: estimate the recorded-date causal response, validate event-time information, and apply the action rule that combines them.

Let the economic event occur at G_i and be recorded at $W_i = G_i + U_i$. Write τ for a mean response indexed by economic onset and m for the same response indexed by the recorded date. A clock law p describes the displacement U . The analyst observes m and a summary $s(p)$ of the clock, then evaluates a query of τ . For a finite set of linear actions, Proposition 2 expresses the requirement in payoff units: each pairwise payoff contrast must admit a common recorded-space bridge across the clock laws sharing the disclosed summary. The resulting bridge determines the action ordering from $(m, s(p))$.

Temporal moments give this calculation in finite-dimensional form. Let $T_r = \sum_h h^r \tau_h$, $M_r = \sum_\ell \ell^r m_\ell$, and $\mu_r = \mathbb{E}[U^r]$. Theorem 1 gives a triangular recursion linking $\mathbf{M}_r$ to $\mathbf{T}_r$ through clock moments up to order r . Theorem 2 compares adjacent clock summaries: within the stated clock and response classes, two environments can share the full recorded response and clock moments through order $r - 1$ and produce different values of T_r . A polynomial timing loss of degree d therefore uses clock moments through degree d ; Theorem 3 maps the moment comparison into scheduled loss and action choice.

Quadratic timing loss gives the scheduling formulas. If $T_0 > 0$ and $\tau_h \geq 0$, the verified-origin scheduling date is the centroid of the latent response, $a_G^* = T_1/T_0 = M_1/M_0 + \mu_1$. Its temporal variance is $V_\tau = V_m - \text{Var}(U)$. With mismatch coefficient γ , a feasible change from recorded-origin execution to verified-origin execution changes minimum scheduled loss by

$$(1) \quad \Delta_{\text{exec}} = \gamma T_0 \text{Var}(U).$$

Flexible provision with cost K is chosen when $K/(\gamma T_0) < V_\tau$. Mean displacement therefore enters the scheduled date, and displacement variance enters the flexibility threshold and the value of the execution origin.

The recorded response requires a causal design of its own. Section 2 defines the clock on a fixed target population, and Proposition 1 recovers m from an interval-anchored comparison design with common conditional untreated changes. Timing heterogeneity

TABLE 1. Decision queries and clock information

Target	Clock information	Recovery equation
Recorded-origin scheduling	Recorded response and execution rule	$a_W^* = M_1/M_0$
Verified-origin scheduling	Mean displacement	$a_G^* = M_1/M_0 + \mu_1$
Verified-origin flexibility	Displacement variance	$V_\tau = V_m - \text{Var}(U)$
Cumulative response	Probability normalization	$T_0 = M_0$
Raw temporal moment T_r	Clock moments through r	Theorem 1
Polynomial timing loss, degree d	Clock moments through d	Theorem 3
Finite linear action ordering	Pairwise common bridges	Proposition 2

$M_r = \sum_\ell \ell^r m_\ell$. The scheduling rows use quadratic mismatch, positive total response mass, and the common-mean clock condition. Finite-window queries also use boundary information.

enters through the association between response and displacement. Under the common-mean clock condition, the population response satisfies $m = A_p \tau$. Section 3 also gives the cross-moment terms that appear when displacement and response magnitude vary jointly.

The European gas application combines these objects around the June 2022 Nord Stream reduction. A June 14 design estimates a three-week response in transit-adjusted cross-border and LNG service as a function of pre-event Nord Stream exposure. Bundesnetzagentur chronology supplies a calendar reference for the first reduction day, and route-flow candidates describe nearby physical origins. The same response profile yields mass, centroid, and temporal variance under these clock choices. A population-clock exercise then maps disclosed displacement moments into scheduling and flexibility decisions. German storage-option contracts provide a callability price and execution window, which give a monetary threshold for quadratic timing curvature. Operator records for Imatra, the Germany–France interruption, and Balticconnector provide a separate documentary timing exercise.

The timing model draws on measurement-error and deconvolution results for convolution, moment recovery, and operator identification (Hall and Lahiri 2008; Schennach 2016, 2019; Carrasco et al. 2007). Functional identification supplies the null-space and bridge representation used for general linear queries (Severini and Tripathi 2006; Santos 2011; Severini and Tripathi 2012). Event-date uncertainty, treatment classification, and staggered-adoption designs concern the causal contrast that enters the clock stage (Ball and Torous 1988; Negi and Negi 2025; Callaway and Sant’Anna 2021; Sun and Abraham 2021; Roth et al. 2023). Policy choice under partial identification supplies the action and regret framework (Manski 2004; Stoye 2009, 2012; Montiel Olea et al. 2026). The appendix collects the operator extensions, clock-action results, sampling calculations, proofs, and empirical details.

2. Environment

2.1. Latent responses and counterfactual queries

Time is discrete and indexed by $t \in \mathbb{Z}$. Unit i has a latent event time $G_i \in \mathbb{Z} \cup \{\infty\}$. The clock stage is defined on units with $G_i < \infty$; units with $G_i = \infty$ can enter the untreated-outcome construction in the causal first stage. For each finite event date g , let $Y_{it}(g)$ denote the potential outcome under an absorbing treatment that begins at g , and let $Y_{it}(\infty)$ denote the untreated potential outcome. For $G_i < \infty$, define the unit-level response at latent horizon h by

$$(2) \quad \tau_{i,h} = Y_{i,G_i+h}(G_i) - Y_{i,G_i+h}(\infty).$$

A cohort-specific dynamic effect is

$$(3) \quad \tau_{g,h} = \mathbb{E}[Y_{i,g+h}(g) - Y_{i,g+h}(\infty) \mid G_i = g].$$

The additive-clock sections use the common mean response $\tau_h = \mathbb{E}[\tau_{i,h} \mid G_i < \infty]$. The operator sections allow a vector or function $\tau \in \mathcal{H}$. The heterogeneous-cohort section returns to $\tau_{g,h}$. A counterfactual query is a functional $Q : \mathcal{X} \rightarrow \mathbb{Z}$ of a latent response path. The operator sections use the continuous linear specialization

$$(4) \quad \psi_a(\tau) = \langle a, \tau \rangle_{\mathcal{H}}.$$

In a finite event window, $\tau \in \mathbb{R}^H$ and $\psi_a = a^\top \tau$. Choices of a produce an impact response, an average over a horizon, a cumulative response, a temporal moment, or a weighted policy loss.

2.2. Clocks and translations

On the finite-event population, a clock is a conditional distribution $K(\cdot \mid G_i, X_i)$ for a recorded timing signal W_i , given latent event time G_i and covariates X_i . A clock has point timing on a maintained support when G_i is measurable with respect to (W_i, X_i) . A clock is fuzzy on that support when a recorded signal corresponds to at least two latent event times with positive conditional probability. The additive benchmark is

$$(5) \quad W_i = G_i + U_i,$$

where U_i is an integer-valued clock error. Positive U_i records the event after its latent onset. Negative U_i records the event before its latent onset. Let $\mathcal{X}$ be a path space closed

under integer translations and define

$$(6) \quad (S_u x)(h) = x(h + u), \quad u \in \mathbb{Z}.$$

The operators satisfy $S_u S_v = S_{u+v}$ and $S_u^{-1} = S_{-u}$. A latent-origin path x^G and its recorded-origin version satisfy

$$(7) \quad x^W = S_U x^G.$$

For a clock law $p_u = \mathbb{P}(U = u)$, define the average clock operator

$$(8) \quad A_p x = \mathbb{E}_p[S_U x] = \sum_{u \in \mathbb{Z}} p_u S_u x$$

whenever the displayed expectation is finite in the response space.

ASSUMPTION 1 (Common-mean clock condition). *On the fixed finite-event target population, $\mathbb{E}[\tau_{i,h} \mid U_i = u] = \tau_h$ for every horizon h and every displacement u in the support, with expectations using the stated analysis weights.*

Assumption 1 requires mean independence of each response coordinate from recording displacement, using fixed target weights. Independence of the entire response vector and U implies this restriction. Conditional variances and higher-order dependence may still vary with U . The condition also allows a nonzero mean displacement and dependence between U and G . Measurement-error models distinguish mean restrictions of this kind from independence of the full latent object (Schennach 2016; Hu and Schennach 2008); Schennach (2019) gives a separate subindependence characterization for convolution. Here the restriction is stated directly for conditional response means. A conditional version applies within covariate or cohort cells, with their weights retained in aggregation. Appendix D gives the mixing equation. Section 3 displays the additional cross-moments when timing is associated with response magnitude.

2.3. A comparison design supplying the causal contrast

Let $\mathcal{E}$ denote the target population with finite event dates. Expectations in the clock stage use this same population and the same aggregation weights at every horizon. The recorded-horizon causal contrast is

$$(9) \quad m_\ell = \mathbb{E}\left[Y_{i,W_{i+\ell}}(G_i) - Y_{i,W_{i+\ell}}(\infty) \mid i \in \mathcal{E}\right].$$

This object requires an untreated-outcome design. An observed outcome profile enters the clock stage as m when that design identifies the displayed contrast. The following result is

a sufficient conditional-comparison construction using the cohort and calendar-time logic of modern difference-in-differences (Callaway and Sant’Anna 2021; Sun and Abraham 2021; Roth et al. 2023). Its role is to supply the input contrast; estimator construction and treatment-comparison inference follow the chosen first-stage design.

PROPOSITION 1 (Interval-anchored causal contrast). *For each target unit, suppose $G_i \in [\underline{G}_i, \bar{G}_i]$ and choose an observed baseline $r_i < \underline{G}_i$. Assume consistency and $Y_{i,r_i}(G_i) = Y_{i,r_i}(\infty)$. Let $\mathcal{C}$ be a comparison population whose observed outcomes equal their untreated potential outcomes at every evaluated calendar pair (r, t) . For pre-event covariates X , define*

$$(10) \quad g_0(x; r, t) = \mathbb{E}[Y_{jt} - Y_{jr} \mid j \in \mathcal{C}, X_j = x].$$

Suppose target covariate support is contained in comparison support at every evaluated pair, and for every target cell (x, w, r) , at $t = w + \ell$,

$$(11) \quad \mathbb{E}[Y_{it}(\infty) - Y_{ir}(\infty) \mid i \in \mathcal{E}, X_i = x, W_i = w, r_i = r] = g_0(x; r, t).$$

With finite displayed expectations and observed outcomes on this support,

$$(12) \quad m_\ell = \mathbb{E}[Y_{i,W_i+\ell} - Y_{i,r_i} - g_0(X_i; r_i, W_i + \ell) \mid i \in \mathcal{E}].$$

PROOF. Comparison consistency identifies g_0 on target support. In each target cell, observed change is the treatment response at t plus the untreated change from r to t , because baseline consistency gives $Y_{ir} = Y_{ir}(\infty)$. Equation (11) identifies the conditional mean of the latter change. Subtract g_0 and average over the target cells to obtain (12). $\square$

An implementation estimates the comparison change at each required calendar pair, predicts it for target covariates, and averages the resulting target residuals with the fixed analysis weights. The comparison-population condition includes treatment status and exposure to spillovers at those dates. Equation (11) states parallel untreated changes within recorded-clock cells; intervals locate an admissible baseline, and the comparison design supplies the untreated change. Anticipatory behavior is accommodated by choosing a baseline at which the stated pre-event consistency condition holds. An announcement and an implementation can also define separate interventions; the intervention represented by G is fixed before selecting its recorded clock.

Under Assumption 1, conditioning the identified contrast on U gives

$$(13) \quad m_\ell = \sum_u p_u \tau_{\ell+u}, \quad m = A_p \tau.$$

Clock estimation or validation supplies information about p ; the comparison design supplies the untreated counterfactual in (9). For a sensitivity analysis, define d_ℓ as the

target-population mean of the untreated change from r_i to $W_i + \ell$ minus its comparison prediction g_0 , and set $\delta_h(u) = \mathbb{E}[\tau_{i,h} \mid U = u] - \tau_h$. The mean observed design score then equals $(A_p \tau)_\ell + d_\ell + \sum_u p_u \delta_{\ell+u}(u)$. Each deviation has a separate identifying restriction and data requirement.

3. Timing Information for Queries

A timing summary records features of the clock law that accompany the recorded response in a query calculation.

DEFINITION 1 (Query-identifying timing summary). *Let $\mathcal{P}$ be a class of clock laws, let $\mathcal{T}$ be a class of response paths, and let Q be a query. A map $s : \mathcal{P} \rightarrow \mathcal{S}$ identifies Q over $\mathcal{P} \times \mathcal{T}$ when, for every $(p, \tau), (p', \tau') \in \mathcal{P} \times \mathcal{T}$,*

$$(14) \quad A_p \tau = A_{p'} \tau' \quad \text{and} \quad s(p) = s(p') \implies Q(\tau) = Q(\tau').$$

3.1. Temporal moments

Assume $\sum_h (1 + |h|^r) |\tau_h| < \infty$ and $\mathbb{E}|U|^r < \infty$. Define

$$(15) \quad T_j(x) = \sum_{h \in \mathbb{Z}} h^j x_h, \quad T_j = T_j(\tau), \quad M_j = T_j(m), \quad \mu_j = \mathbb{E}[U^j],$$

with $T_0 = \sum_h \tau_h$, $M_0 = \sum_\ell m_\ell$, and $\mu_0 = 1$. Collect moments through order r as $\mathbf{T}_r = (T_0, \dots, T_r)^\top$ and $\mathbf{M}_r = (M_0, \dots, M_r)^\top$. For $j, k \in \{0, \dots, r\}$, define the Pascal-action matrix

$$(16) \quad [\Pi_r(u)]_{jk} = \begin{cases} \binom{j}{k} (-u)^{j-k}, & k \leq j, \\ 0, & k > j. \end{cases}$$

The shift identity is

$$(17) \quad \mathbf{T}_r(S_u x) = \Pi_r(u) \mathbf{T}_r(x).$$

The following recursion is the additive measurement-error moment calculation under the sign convention $W = G + U$. Hall and Lahiri (2008, Section 2.2) gives the integer-moment recovery formula in a deconvolution model. Here the recursion establishes the sufficiency side of the timing-summary comparison.

THEOREM 1 (Moment transport). *For every integer $r \geq 0$,*

$$(18) \quad \mathbf{M}_r = \mathbb{E}[\Pi_r(U)] \mathbf{T}_r.$$

Its r -th component is

$$(19) \quad M_r = \sum_{k=0}^r \binom{r}{k} (-1)^k \mu_k T_{r-k}.$$

Hence m and $s_r(p) = (\mu_1, \dots, \mu_r)$ determine $T_0, \dots, T_r$ recursively; set $s_0(p)$ equal to the empty tuple.

The online appendix gives the moment-interchange argument and the recursion.

THEOREM 2 (Moment-order minimality). *Fix $r \geq 1$. Suppose the clock class contains two bounded-support laws p and p' with*

$$(20) \quad \mu_j(p) = \mu_j(p') \quad \text{for } j = 1, \dots, r-1, \quad \mu_r(p) \neq \mu_r(p'),$$

and suppose the response class contains every nonnegative finite-support path with total mass one. Then there exist response paths τ, τ' in the class such that

$$(21) \quad A_p \tau = A_{p'} \tau', \quad s_{r-1}(p) = s_{r-1}(p'), \quad T_r(\tau) \neq T_r(\tau').$$

The online appendix constructs the two environments by exchanging reflected clock laws and using convolution commutativity. Theorem 2 compares adjacent summaries s_{r-1} and s_r on the displayed clock and response classes. Its scope is the raw-moment hierarchy; a different query or a restricted response class has its own information requirement. Choudhary and Mitra (2018) characterize rank-two null-space ambiguity for blind convolution; the associated working paper develops almost-everywhere pair ambiguity (Choudhary and Mitra 2014, Section IV). Those comparisons allow the clock's lower moments to change. Here the construction enforces their equality through $r-1$ and changes the selected r -th response moment. Commutativity is the shared algebraic ingredient; the preserved summaries specify the information comparison. Corollary 2 gives summaries for the mean and variance queries.

COROLLARY 1 (Clock conservation). *If $\tau \in \ell^1(\mathbb{Z})$, then*

$$(22) \quad \sum_{\ell \in \mathbb{Z}} m_\ell = \sum_{h \in \mathbb{Z}} \tau_h.$$

COROLLARY 2 (Temporal centroid and dispersion). *The first two causal moments satisfy*

$$(23) \quad T_1 = M_1 + \mu_1 M_0,$$

$$(24) \quad T_2 = M_2 + 2\mu_1 T_1 - \mu_2 M_0.$$

If $\tau_h \geq 0$ and $T_0 > 0$, define the response-mass distribution $q_h = \tau_h/T_0$ and the recorded-mass distribution m_ℓ/M_0 . Their variances V_τ and V_m satisfy

$$(25) \quad \frac{T_1}{T_0} = \frac{M_1}{M_0} + \mu_1, \quad V_\tau = V_m - \text{Var}(U), \quad V_\tau = \frac{T_2}{T_0} - \left(\frac{T_1}{T_0}\right)^2.$$

The clock summaries are μ_1 for mean location and $\text{Var}(U)$ for temporal variance.

Full-axis moment statements require the displayed absolute-moment conditions. A persistent level response is handled through a specified finite-window target or an integrable response transformation. For a finite window, the recorded total weights each latent horizon by the probability that its translated coordinate remains inside the window; Online Appendix A gives that identity. The observation domain and its boundary information therefore belong to the query specification.

3.2. Queries with response-dependent recording

Let $T_{ji} = \sum_h h^j \tau_{i,h}$, $T_j = \mathbb{E}[T_{ji}]$, and $M_j = \sum_\ell \ell^j m_\ell$ on the same target population. The first-order pathwise identity can be averaged whenever $\mathbb{E}[\sum_h (1 + |h| + |U_i|)|\tau_{i,h}|] < \infty$, giving

$$(26) \quad M_0 = T_0, \quad T_1 = M_1 + \mu_1 M_0 + \text{Cov}(U_i, T_{0i}).$$

The total is invariant to the association between displacement and response. For the first moment, mean-displacement correction uses the restriction $\text{Cov}(U_i, T_{0i}) = 0$; Assumption 1 supplies this restriction as part of its pathwise conditional-mean requirements. Delays associated with the total size of the response contribute the covariance in (26). Information about that joint association can be included explicitly in the query equation.

$$(27) \quad T_2 = M_2 + 2\mu_1 T_1 - \mu_2 T_0 + 2\text{Cov}(U_i, T_{1i}) - \text{Cov}(U_i^2, T_{0i}).$$

Equation (27) uses the corresponding second-order absolute cross-moments. Online Appendix A derives the general identity, and Online Appendix D gives a two-cell example. These equations distinguish information about the marginal clock law from information about recording and response jointly. The gas application evaluates their measurement-path counterparts under its stated candidate-date weights.

4. Decisions and Timing Information

Let a finite action set $\mathcal{A}$ have losses $L_a(\tau) = k_a + \langle q_a, \tau \rangle$. For recorded response m and clock summary s , let $\mathcal{E}(m, s)$ collect every (τ, p) in the maintained response and clock

classes satisfying $m = A_p \tau$ and $s(p) = s$. A decision is identified when its optimal-action correspondence agrees across this environment fiber.

PROPOSITION 2 (Decision fiber and common bridges). *The optimal-action correspondence is identified at (m, s) exactly when $\arg \min_{a \in \mathcal{A}} L_a(\tau)$ is the same for every $(\tau, p) \in \mathcal{E}(m, s)$. A sufficient condition is that for every action pair (a, b) there is a recorded-space vector $v_{ab,s}$ satisfying*

$$(28) \quad A_p^* v_{ab,s} = q_a - q_b \quad \text{for every clock law } p \text{ with } s(p) = s.$$

Under this condition, $L_a(\tau) - L_b(\tau) = k_a - k_b + \langle v_{ab,s}, m \rangle$, so (m, s) determines the action ordering. If two environments in one fiber reverse the sign of a pairwise variable-loss contrast, an interval of relative fixed costs in the associated two-action problem produces different optimal actions.

Proposition 2 applies the linear-functional representation in Section 6 to payoff contrasts. Components shared by every q_a cancel from those contrasts. Online Appendix J gives the proof and the policy constructions used below.

Let $c(z) = \sum_{j=0}^d c_j z^j$ be a nonnegative polynomial on $\mathbb{R}$, with even degree $d \geq 2$ and $c_d > 0$. A scheduled action a belongs to a compact set $\mathcal{A} \subset \mathbb{R}$, and flexible provision costs $K \geq 0$. Define

$$(29) \quad L_c(a; \tau) = \sum_h \tau_h c(h - a), \quad L_c(F; \tau) = K.$$

THEOREM 3 (Decision-order moment requirement). *Suppose $\tau_h \geq 0$, $T_0 > 0$, the moments through order d are finite, and Assumption 1 holds. The recorded response m and clock moments $s_d(p)$ identify $a \mapsto L_c(a; \tau)$, its minimum on $\mathcal{A}$, and the comparison with K . Under the clock and response classes of Theorem 2, suppose two clock laws agree through moment $d - 1$ and differ at moment d . There exist two environments with the same recorded response and the same s_{d-1} for which minimum scheduled losses differ. Every K strictly between these minima yields different scheduled-versus-flexible choices.*

The proof expands $c(h - a)$ in temporal moments and applies Theorems 1 and 2; Online Appendix J gives the calculation.

For the quadratic specialization, let $\tau_h \geq 0$ have $T_0 > 0$ and a finite second moment. A scheduled program incurs mismatch loss with coefficient $\gamma > 0$, and flexible provision matches the profile at cost $K \geq 0$:

$$(30) \quad L(a; \tau) = \gamma \sum_h \tau_h (h - a)^2, \quad L(F; \tau) = K, \quad \lambda = \frac{K}{\gamma T_0}.$$

The date a is measured from an operational origin available to the execution system.

PROPOSITION 3 (Scheduling and flexibility). Write $\bar{h} = T_1/T_0$. Scheduled loss satisfies

$$(31) \quad \frac{L(a; \tau)}{\gamma T_0} = V_\tau + (a - \bar{h})^2, \quad a^* = \bar{h}, \quad \frac{\min_a L(a; \tau)}{\gamma T_0} = V_\tau.$$

Flexible provision is strictly preferred when $\lambda < V_\tau$, scheduling is strictly preferred when $\lambda > V_\tau$, and both are optimal at equality. Under Assumption 1,

$$(32) \quad a^* = \frac{M_1}{M_0} + \mu_1, \quad V_\tau = V_m - \text{Var}(U).$$

Mean displacement determines the verified-origin date and displacement variance determines the flexibility threshold. Online Appendix J gives policy witnesses associated with the first two moment orders.

Execution at the recorded origin uses the response m directly. For nonnegative unit profiles with finite absolute cross-moments,

$$(33) \quad \mathbb{E} \left[\sum_h \tau_{i,h} \{h - (U_i + a)\}^2 \right] = \sum_\ell m_\ell (\ell - a)^2.$$

The optimal recorded-origin offset is M_1/M_0 . Write L_W^* and L_G^* for minimum scheduled losses under recorded-origin and verified-origin execution. Under Assumption 1,

$$(34) \quad L_W^* = \gamma T_0 V_m, \quad L_G^* = \gamma T_0 V_\tau, \quad \Delta_{\text{exec}} = L_W^* - L_G^* = \gamma T_0 \text{Var}(U).$$

This comparison holds the need profile and loss specification fixed. An execution-system investment can be evaluated by comparing its cost with Δ_{exec} . Validation information affects the decision through the admissible clock summaries.

Partial validation supplies a set of admissible profiles. With loss divided by γT_0 , deterministic minimax regret is

$$(35) \quad R^*(\lambda; \mathcal{H}) = \inf_{d \in \mathbb{R} \cup \{F\}} \sup_{(\bar{h}, \nu) \in \mathcal{H}} \{ \ell_d(\bar{h}, \nu) - \min(\lambda, \nu) \}, \quad \ell_a = \nu + (a - \bar{h})^2, \quad \ell_F = \lambda.$$

Online Appendix J derives the closed-form solution for the moment regions used in the application.

5. Clock Actions and Nonlinear Queries

The temporal-moment results concern population averages. For a realized path, query recovery follows the action of one displacement. Orbit notation collects the latent paths compatible with a recorded path and displacement support. First-passage equivariance

and the observations entering or leaving a finite burden window then give concrete measurement implications.

DEFINITION 2 (Clock action). *A query $Q : \mathcal{X} \rightarrow \mathbb{Z}$ admits a clock action when a group action $\{\rho_u : u \in \mathbb{Z}\}$ satisfies*

$$(36) \quad Q(S_u x) = \rho_u(Q(x))$$

for every path and displacement in the maintained translation-stable domain.

For a continuous linear vector query q with matrix action R_q , averaging gives $q(A_p \tau) = \mathbb{E}_p[R_q(U)]q(\tau)$ whenever the expectation and linear map commute. Temporal moments use the Pascal matrix in (16). Nonlinear queries below retain the realized translation and its support.

The population relation (13) averages clock displacements. Pathwise analysis keeps the realized displacement. For an observed path x^W and displacement support $\Gamma \subseteq \mathbb{Z}$, define the latent-path orbit

$$(37) \quad \mathcal{O}_\Gamma(x^W) = \{S_{-u}x^W : u \in \Gamma\}.$$

THEOREM 4 (Clock-orbit identification). *Suppose $x^G \in \mathcal{X}$, $x^W = S_U x^G$, and $U \in \Gamma$. If Q admits the clock action in Definition 2, then*

$$(38) \quad \mathcal{I}_Q(x^W, \Gamma) = Q(\mathcal{O}_\Gamma(x^W)) = \{\rho_{-u}(Q(x^W)) : u \in \Gamma\}.$$

PROOF. For each $u \in \Gamma$, the latent path associated with that displacement is $S_{-u}x^W$. Apply (36) to each path in $\mathcal{O}_\Gamma(x^W)$. $\square$

A translation-invariant query has ρ_u equal to the identity action on its value space. For linear queries on $\ell^1(\mathbb{Z})$, this property can be read from the loading sequence.

PROPOSITION 4 (Shift-invariant linear queries). *Let $S \subseteq \mathbb{Z}$ be a set of admissible deterministic clock errors, and let $L_a(x) = \sum_h a_h x_h$ for $a \in \ell^\infty(\mathbb{Z})$. The value $L_a(S_u x)$ is the same for every $u \in S$ and every $x \in \ell^1(\mathbb{Z})$ exactly when*

$$(39) \quad a_{j+u} = a_{j+v} \quad \text{for every } j \in \mathbb{Z} \text{ and } u, v \in S.$$

If the subgroup generated by $S - S$ equals $\mathbb{Z}$, then a_h is constant across h , and L_a is a scalar multiple of full-axis cumulative response.

5.1. First passage

Let $x = (r, y)$ contain a replacement path $r(h)$ and an activity path $y(h)$. Let $\bar{\mathbb{Z}} = \mathbb{Z} \cup \{-\infty, +\infty\}$, extend integer translation by $(\pm\infty) - u = \pm\infty$, and set $\inf \emptyset = +\infty$. For $\zeta = (q, p, \kappa)$, define the unrestricted first-passage query with values in $\bar{\mathbb{Z}}$

$$(40) \quad D_\zeta(x) = \inf \{h \in \mathbb{Z} : r(h+s) \geq q, y(h+s) \geq \kappa, s = 0, \dots, p-1\}.$$

The infimum is taken on a translation-stable observation domain; finite-window reporting is handled after this query is formed.

COROLLARY 3 (First-passage clock action). *For every integer u ,*

$$(41) \quad D_\zeta(S_u x) = D_\zeta(x) - u.$$

Hence $\rho_u(d) = d - u$, and Theorem 4 gives

$$(42) \quad \mathcal{I}_{D_\zeta}(x^W, \Gamma) = D_\zeta(x^W) + \Gamma.$$

If $D_\zeta^W = D_\zeta(x^W)$ and $D_\zeta^G = D_\zeta(x^G)$ are integrable across units, then

$$(43) \quad \mathbb{E}[D_\zeta^G] = \mathbb{E}[D_\zeta^W] + \mathbb{E}[U].$$

For finite first-passage values, a pair of endpoint definitions ζ and ζ' yields an origin-free difference:

$$(44) \quad D_{\zeta'}(S_u x) - D_\zeta(S_u x) = D_{\zeta'}(x) - D_\zeta(x).$$

Equation (44) supplies a crossed clock-by-query restriction. A nonzero mixed difference across two recorded origins records changes in measurement ingredients beyond a pure origin shift.

5.2. Finite-horizon burden

Let $b_x(h) \in [0, B]$ be a translation-covariant shortfall path, so $b_{S_u x}(h) = b_x(h + u)$, and define

$$(45) \quad \Xi_H(x) = \sum_{h=0}^H b_x(h).$$

PROPOSITION 5 (Boundary sensitivity of cumulative burden). *For every integer u ,*

$$(46) \quad \Xi_H(S_u x) - \Xi_H(x) = \begin{cases} \sum_{j=1}^u b_x(H+j) - \sum_{j=0}^{u-1} b_x(j), & u > 0, \\ 0, & u = 0, \\ \sum_{j=u}^{-1} b_x(j) - \sum_{j=H+u+1}^H b_x(j), & u < 0. \end{cases}$$

Consequently,

$$(47) \quad |\Xi_H(S_u x) - \Xi_H(x)| \leq B|u|.$$

For the normalized burden $\bar{\Xi}_H = \Xi_H/(H+1)$,

$$(48) \quad |\bar{\Xi}_H(S_u x) - \bar{\Xi}_H(x)| \leq \frac{B|u|}{H+1}.$$

If $\sum_{h \in \mathbb{Z}} b_x(h) < \infty$, the full-axis burden $\Xi_{\mathbb{Z}}(x) = \sum_{h \in \mathbb{Z}} b_x(h)$ satisfies $\Xi_{\mathbb{Z}}(S_u x) = \Xi_{\mathbb{Z}}(x)$.

5.3. Several records for one event

Several recorded clocks can be reconciled by requiring their reindexed paths to agree for a common latent origin. Online Appendix B gives the joint orbit, its query image, and the compatibility equations. The target intervention, physical path, endpoint, and observation domain remain fixed throughout this comparison.

6. Linear Queries under a Clock Operator

The linear-functional identification literature provides the following characterization (Severini and Tripathi 2006; Santos 2011; Severini and Tripathi 2012; Carrasco et al. 2007). Let $A : \mathcal{H} \rightarrow \mathcal{Y}$ be bounded, let $m = A\tau$, and consider $\langle a, \tau \rangle$. The query is constant on each observational fiber when $a \in \ker(A)^\perp = \overline{\text{Range}(A^*)}$. A bounded linear representation in the observed response has the form

$$(49) \quad \langle a, \tau \rangle = \langle b, m \rangle, \quad A^* b = a,$$

and exists exactly when $a \in \text{Range}(A^*)$. Online Appendix C states the characterization and its proof. This operator geometry supports the timing calculation; Theorem 2 addresses the additional information question when the clock law itself is described through nested moment summaries.

Appendix C records the spectral specialization and generalized least-squares bridge (Zinde-Walsh 2014). Appendix E applies differentiable plug-in inference when a validation sample estimates the clock; constant-rank pseudoinverse calculus supplies its bridge path (Golub and Pereyra 1973). It separates first-order compatibility from local query identification and retains both in its covariance calculation. Appendices F and G apply Gaussian projection and identified-set constructions. These calculations supply the operator and sampling implementation of a specified query.

7. Timing Information in European Gas Data

The application uses European gas data around the June 2022 Nord Stream reduction to connect an estimated response, alternative event-time origins, and timing-sensitive decisions. ENTSOG provides daily network flows, Eurostat provides pre-event gas balances and exposure measures, and public regulator and operator records provide event dates. The causal and clock calculations use fixed target definitions within each exercise.

7.1. A network response and its event-time clock

Let N_{it} be transit-adjusted cross-border and LNG service divided by 2021 baseline service, and let E_i be pre-event Nord Stream exposure. The local design estimates

$$(50) \quad N_{it} = \alpha_i + \delta_t + \beta E_i \mathbf{1}\{t \geq \text{June 14, 2022}\} + g(\text{weather}_{it}) + \varepsilon_{it}.$$

Under exposure-specific parallel counterfactual changes, predetermined exposure and scale, and consistency of the measured network-service outcome, β is the June 14 network-reallocation gradient. Inference uses fifteen exposure units.

TABLE 2. June 14 network-reallocation contrast

Window	Effect	Delete-one 95% interval	Familywise exact p	Lead joint p
21 days	25.5 pp	[11.5, 39.5]	0.0272	0.966
28 days	27.7 pp	[15.1, 40.2]	0.0086	0.966

Effects are percentage points of baseline daily service for a 10-percentage-point increase in pre-event Nord Stream exposure. Delete-one intervals use a market jackknife. Familywise exact score inference uses the maximum statistic across the two displayed windows and enumerates their shared cluster-score sign patterns under null sign symmetry. The weekly lead test covers event weeks -3 and -2 , with week -1 omitted.

The 21-day estimate is 25.53 percentage points and the 28-day estimate is 27.68 points. A shared-sign maximum-statistic test over the two windows gives $p = 0.01$, and the weekly lead joint p -value is 0.97. Alternative scaling choices retain positive estimates in both windows; Online Appendix I reports the specification and influence calculations.

Let $\widehat{\beta}_k$ be the exposure interaction for event week k , relative to week -1 . Define the recorded three-week response profile by

$$(51) \quad \widehat{m}_\ell^N = \widehat{\beta}_k, \quad \ell = 7k, \dots, 7k + 6, \quad k = 0, 1, 2, \quad \widehat{m}_\ell^N = 0 \text{ outside days } 0, \dots, 20.$$

Its mass, centroid, and variance are

$$(52) \quad \widehat{M}_0^N = \sum_\ell \widehat{m}_\ell^N, \quad \widehat{a}_W^N = \frac{\sum_\ell \ell \widehat{m}_\ell^N}{\widehat{M}_0^N}, \quad \widehat{V}_W^N = \frac{\sum_\ell (\ell - \widehat{a}_W^N)^2 \widehat{m}_\ell^N}{\widehat{M}_0^N}.$$

TABLE 3. Three-week response profile

Event week	Daily response	Cluster- t_{14} 95% interval	Delete-one range
0 (days 0–6)	11.58 pp	[-5.45, 28.61]	[-0.04, 14.49]
1 (days 7–13)	27.43 pp	[10.92, 43.94]	[15.88, 29.32]
2 (days 14–20)	34.62 pp	[11.39, 57.85]	[20.15, 39.96]

Each coefficient is constant within its seven-day bin in (51). Cluster intervals use a Student t critical value with 14 degrees of freedom. Delete-one columns re-estimate the weekly profile after removing one market.

Bundesnetzagentur’s 2022 monitoring report records Nord Stream 1 at about 60 percent of maximum capacity on June 14 and 40 percent on June 15 (Bundesnetzagentur and Bundeskartellamt 2022). June 14 therefore supplies the regulator reference for the first reduction day. A route-flow screen around that date gives June 13 and June 16 gas-day candidates, corresponding to $U = W - G \in \{1, -2\}$. For each deterministic candidate,

$$(53) \quad a_G^N = \widehat{a}_W^N + U, \quad G + a_G^N = W + \widehat{a}_W^N, \quad V_G^N = \widehat{V}_W^N.$$

The regulator reference sets $U = 0$. The route candidates give a finite sensitivity set for physical onset.

TABLE 4. Clock queries from the response profile

Query	Point or clock set	Sampling interval / outer set	Delete-one range
Three-week response mass	515.4 pp-days	[132.2, 898.6]	[251.9, 584.8]
Recorded/regulator centroid	12.19 days	[10.70, 13.68]	[11.71, 13.93]
Physical-candidate centroid	[10.19, 13.19] days	[8.70, 14.68]	[9.71, 14.93]
Pathwise temporal variance	25.95 day ²	[17.47, 34.42]	[11.97, 27.45]

Mass uses percentage-point days per ten-percentage-point exposure. Centroids use days relative to the displayed origin. The physical-centroid row takes the outer union over the two route candidates. Variance is invariant across deterministic translations.

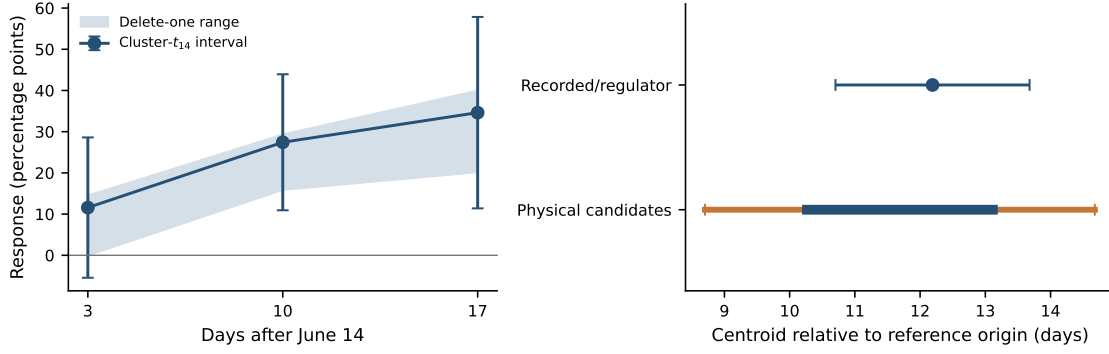


FIGURE 1. Network response and event-time translation

The left panel gives weekly coefficients and sampling intervals. The right panel gives the regulator-referenced centroid and the route-flow sensitivity set. Each origin maps to the same calendar centroid through (53).

The estimated mass is 515.43 percentage-point days and the regulator-referenced centroid is 12.19 days. The route candidates give a physical-origin centroid range of 10.19–13.19 days. The corresponding sampling intervals are reported in Table 4.

7.2. Population timing information and policy loss

For the policy exercise, let R_{it} be allocated Russian service and let A_{it}^j be the signed observed-minus-baseline deviation for external supply, production, or storage withdrawal. Define the projected residual service need

$$(54) \quad Q_{it}^R = (\widehat{R}_{it}^0 - R_{it})_+, \quad A_{it}^\Sigma = \sum_{j \in J} A_{it}^j, \quad b_{it} = \frac{(Q_{it}^R - A_{it}^\Sigma)_+}{B_{it}},$$

where B_{it} sums the component baselines. The seven markets with complete 2022–2024 paths define the common target used in the population-clock calculation. The candidate displacement law is applied to that windowed target, yielding $m = A_p \tau$ under the common-mean construction. Online Appendix I gives the accounting and clock specification.

The first three disclosure levels recover

$$(55) \quad T_0 = M_0, \quad T_1 = M_1 + \mu_1 M_0, \quad T_2 = M_2 + 2\mu_1 T_1 - \mu_2 M_0.$$

Cumulative mass is 172.85 share-days. Support information gives a scheduled-date range of 218.9–491.9 days; mean displacement fixes the date at 399.0. Variance disclosure fixes the flexibility threshold at 92.29 percent of V_m . Maximum normalized regret moves from 17.71 percent under support information to 7.55 percent after mean disclosure and reaches zero after variance disclosure. In the same clock construction, verified-origin

TABLE 5. Timing information for scheduling and flexibility

Clock information	Scheduled date a^*	V_τ/V_m	$\sup_{\lambda \geq 0} R^*/V_m$
Support	[218.9, 491.9]	[82.3, 100.0]%	17.71%
+ mean	399.0	[84.1, 99.2]%	7.55%
+ variance	399.0	92.3%	0.00%

The rows use one recorded target and one candidate displacement law. Dates use days. Variance and regret are percentages of the observed V_m . The first two rows use moment-compatible regions conditional on the target and displacement support.

execution changes minimum scheduled loss by 7.71 percent of the recorded-origin benchmark.

7.3. A storage-option benchmark

German storage options provide a price and execution window for the flexibility comparison. The second 2022 SSBO tender contracted 35.7175 TWh, including 7.1435 TWh callable from October 2022 through January 2023 (Trading Hub Europe 2022a). Weighting the awarded bids by lot counts gives EUR 232.114 million in storage fees and EUR 249.130 million in availability fees; the latter equals EUR 34.88 per callable MWh (Trading Hub Europe 2022b). The award announcement reports approximately EUR 481 million in combined fees (Trading Hub Europe 2022c). Contract terms assign the availability charge to continuous call-off capacity and price commodity delivery separately (Trading Hub Europe 2022d).

ENTSOE assigns Germany a disruption value of EUR 114.1 per MWh for project appraisal (European Network of Transmission System Operators for Gas 2020). Applying the callable quantity uniformly across the 123-day contract window covers 6.31 TWh of the projected residual path. The corresponding gross covered-disruption value is EUR 720.44 million, and combined contract fees imply EUR 76.22 per covered MWh.

TABLE 6. German storage-option benchmark

Quantity	Value
Covered projected shortfall	6.314 TWh
ENTSOE German CoDG	EUR 114.1/MWh
Gross covered-disruption value	EUR 720.4 million
Combined-fee break-even	EUR 76.22/MWh
CoDG headroom after combined fees	EUR 37.88/MWh

The calculation uses the projected residual path, uniform daily callability over October 1, 2022 through January 31, 2023, and ENTSOE's German disruption value. Commodity call charges, storage constraints, realized withdrawals, and incidence enter a welfare calculation through additional inputs.

The covered contract-window profile has centroid 59.79 days from October 1 and variance 1199.83 squared days. Under the quadratic benchmark, a scheduled plan has minimum loss gQ_0V_q . Equating that loss to the EUR 249.13 million availability charge gives

$$(56) \quad g_{\text{avail}}^* = \frac{F_{\text{avail}}}{Q_0V_q} = \text{EUR } 0.0329 \text{ per MWh per squared day.}$$

This value is the timing-curvature threshold implied by the contract price for the specified need profile.

7.4. Documentary timing checks

Public operator records provide event-date checks on named interruptions. Gasum and Gasgrid document the Imatra interruption, GRTgaz records the Germany–France physical interruption, and Gasgrid records the Balticconnector pressure-loss and isolation episode (Gasum 2022; Gasgrid Finland 2022; GRTgaz 2022; Gasgrid Finland 2023). Within the 2021–2024 Gasgrid English-bulletin frame and the retained ENTSOG routes, the eligible abrupt shutdown events are Imatra and Balticconnector. The reference route-flow rule matches both civil dates. Calendar-date precision maps each event to the adjacent compatible gas-day labels. Online Appendix I states the screening rule and reports the associated sensitivity calculations.

8. Conclusion

A decision and its execution origin determine which features of event timing enter evaluation. Under additive clock displacement, moments of the recorded response and the clock law recover the corresponding temporal moments of the latent response. A polynomial timing loss of degree d therefore uses clock moments through degree d . The adjacent-moment construction shows that the next clock moment can change scheduled loss even when the recorded response and all lower-order clock moments are held fixed.

For quadratic scheduling, mean displacement determines the verified-origin date and displacement variance determines the flexibility threshold. Under the common-mean clock condition, the same variance gives the scheduled-loss difference between recorded-origin and verified-origin execution. The gas application uses three separate empirical inputs. Exposure-specific parallel counterfactual changes identify the recorded-date network-reallocation gradient around the June 2022 Nord Stream reduction. Regulator and route records provide a finite set of event-time origins. The clock law and quadratic action rule then translate these inputs into scheduling and storage-option quantities. The online appendix contains the proofs, operator extensions, clock-action results, sampling calculations, and empirical specifications.

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